

Final Report: Marketing Campaign Business Insights Report (EDA)

SkillCraft Technology – Task 04

Repository: SCT_DA_4

Executive Summary

This project focused on performing Exploratory Data Analysis (EDA) on a marketing campaign performance dataset to evaluate campaign effectiveness, customer funnel efficiency, and channel-specific profitability. The primary objective was to understand the drivers behind marketing performance rather than develop predictive machine learning models.

Using Python for data cleaning and KPI generation, followed by Power BI for dashboard creation, this project identified the most effective marketing channels, uncovered inefficiencies in customer acquisition funnels, and provided strategic recommendations for marketing budget optimization.

Project Objectives

- Clean and preprocess raw campaign data
 - Calculate critical marketing KPIs
 - Evaluate campaign performance across channels
 - Visualize marketing funnel stages
 - Analyze revenue, cost, and ROI relationships
 - Build an executive Power BI dashboard
 - Recommend budget allocation strategies based on ROI
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Dataset Description

The dataset consisted of 10,000 marketing campaign records with the following features:

- CampaignID
- StartDate
- EndDate
- Channel
- Impressions
- Clicks

- Leads
- Conversions
- Cost_USD
- Revenue_USD

These variables enabled comprehensive analysis of customer acquisition funnels, financial performance, and strategic marketing efficiency.

Data Cleaning & Preparation

Several preprocessing steps were performed to ensure data quality:

Data Cleaning Steps:

- Removed duplicate entries
- Handled missing values
- Standardized date formats
- Validated cost and revenue fields
- Eliminated invalid numerical records
- Created derived analytical columns

KPI Metrics Created:

- Click-Through Rate (CTR)
- Conversion Rate
- Lead Rate
- Cost Per Click (CPC)
- Cost Per Acquisition (CPA)
- Return on Investment (ROI)
- Campaign Duration

This process transformed raw marketing data into a business intelligence-ready dataset.

Exploratory Data Analysis (EDA)

Channel Performance Analysis

Each marketing channel was evaluated based on:

- Revenue generation
- Marketing spend

- ROI
- CPA
- Conversion efficiency

Major Findings:

- Google Ads generated the highest ROI and strongest profitability.
 - Facebook consumed significant budget but underperformed relative to spend.
 - Email campaigns achieved the lowest CPA and highly efficient conversions.
 - Instagram displayed strong visibility but weaker final conversion rates.
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Funnel Analysis

The customer acquisition funnel was structured as:

Impressions → Clicks → Leads → Conversions

Key Observations:

- Significant drop-off between clicks and leads
- Indicates potential landing page inefficiencies
- Highlights opportunities for improved lead nurturing
- Suggests conversion optimization opportunities

Business Interpretation:

- Weak ad creatives reduce CTR
 - Landing page performance affects lead generation
 - Sales process impacts final conversions
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Dashboard Development

A professional Power BI dashboard was developed to provide interactive business insights.

Dashboard Features:

- Total Spend KPI
- Total Revenue KPI
- Total Conversions KPI
- Average ROI KPI
- Average CTR KPI
- Funnel Performance Chart

- Revenue by Channel
 - ROI by Channel
 - Monthly Trends Analysis
 - Efficiency Scatter Plot
 - Performance Summary Table
 - Executive Recommendations
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Key Insights

- Google Ads delivered the highest ROI, making it the most profitable channel.
 - Facebook demonstrated high spending with lower relative returns.
 - Funnel inefficiencies between clicks and leads indicate landing page optimization needs.
 - Email marketing provided the best cost efficiency and acquisition performance.
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Strategic Recommendations

Budget Allocation:

- Increase budget for Google Ads
- Expand Email Marketing campaigns
- Optimize Instagram targeting before expansion
- Reduce or restructure Facebook advertising spend

Funnel Optimization:

- Improve landing page design
 - Strengthen lead capture systems
 - Enhance campaign targeting strategies
 - Focus on ROI-driven resource allocation
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Business Impact

This project demonstrates practical business intelligence capabilities by:

- Transforming raw data into strategic insights
- Supporting executive decision-making
- Improving marketing budget efficiency
- Identifying operational bottlenecks

- Enhancing customer acquisition performance
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Tools & Technologies Used

- Python (Pandas, NumPy)
 - Jupyter Notebook
 - Excel
 - Power BI
 - GitHub
 - PDF Reporting
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Repository Deliverables

SCT_DA_4/

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|— cleaned_marketing_campaign_data.csv

|— Marketing Campaign Performance Dashboard.pbix

|— dashboard_screenshot.png

|— final_report.pdf

|— analysis.ipynb

└─ README.md

Conclusion

This project successfully converted raw marketing performance data into actionable business intelligence through structured EDA, KPI generation, dashboard visualization, and strategic reporting.

By focusing on ROI, funnel performance, and budget allocation, the analysis provides clear recommendations for improving marketing efficiency and maximizing profitability.

The project serves as a professional portfolio demonstration of:

- Business Analytics
- Marketing Intelligence
- Dashboard Design
- Executive Reporting
- Strategic Decision Support

Final Statement

The SCT_DA_4 project represents a comprehensive end-to-end business analytics solution designed to support data-driven marketing optimization and executive-level strategic planning.